A VERTEX-SPREAD BLOW-UP LEMMA

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Abstract. For graphs H and G, a probability distribution on embeddings $\varphi : H \hookrightarrow G$ is q-vertex-spread if the probability of any prescribed images for distinct source vertices is at most the corresponding power of q. We prove a vertex-spread version of the Blow-up Lemma. Every bounded-degree graph that is compatible with a lower super-regular blow-up instance with candidacy graphs admits a compatible embedding distribution that is $O(1/n)$-vertex-spread, where n is the cluster scale.

1. Introduction

The Blow-up Lemma of Komlós, Sárközy, and Szemerédi is a central tool for embedding bounded-degree spanning graphs into dense host graphs. Roughly speaking, it allows the complete pairs of a blow-up to be replaced by super-regular pairs without losing any bounded-degree spanning subgraph.

In this paper, we prove a probabilistic strengthening of the Blow-up Lemma. Rather than producing one embedding, our result produces a distribution on embeddings in which prescribing the images of s vertices has probability at most C^s/n^s. We also allow a candidate set for the image of each vertex. The precise definitions and the statement of the main theorem are given in Section 2.

Remark. This proof was completed in the Summer of 2024 with no AI assistance. Initially, the theorem was part of a manuscript proving a robust version of the Pósa–Seymour Conjecture. I am still working on completing that application, but I've decided to split the manuscript into pieces and not submit this portion for publication. Independently, Nenadov and Pham [11] and Allen, Böttcher, Kohayakawa, and Neve [1] also obtained vertex-spread versions of the Blow-up lemma, with different proofs.

2. Definitions and the main theorem

We begin with the definition of vertex-spreadness and the standard regularity terminology needed to state the main theorem.

Definition 2.1. For $q \in [0,1]$, finite sets X and Y, and $U \subseteq Y$, a probability distribution μ on functions $f : X \to Y$ is *q-vertex-spread on U* if $\mu(\{f : f(x_i) = y_i \ \forall i \in [s]\}) \leq q^s$ for every two sequences of distinct $x_1, \dots, x_s \in X$ and (not necessarily distinct) $y_1, \dots, y_s \in U$. If μ is q-vertex-spread on Y, then we simply say it is *q-vertex-spread*. For graphs H and G, a probability distribution μ on embeddings $\varphi : H \hookrightarrow G$ is *q-vertex-spread* if $\mu : V(H) \to V(G)$ is q-vertex-spread.

Definition 2.2. For a graph G and two disjoint subsets $U, W \subseteq V(G)$:

- $G[U,W]$ is the bipartite graph with parts U and W induced by G;
- $e_G(U,W) \coloneqq |E(G[U,W])|$; and
- $d_G(U,W) \coloneqq e_G(U,W)/(|U||W|)$ is the *density* of the pair U, W in G.

Definition 2.3. For $\varepsilon > 0$ and $d \in (0,1)$, a bipartite graph G with bipartition (V_1, V_2) is

- *(ε, d)-regular* if $d_G(U,W) = d \pm \varepsilon$ for every $U \subseteq V_1, W \subseteq V_2$ satisfying $|U| > \varepsilon|V_1|$, $|W| > \varepsilon|V_2|$;

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- (ε, d)-*super-regular* if it is (ε, d)-regular and $|N_G(v) \cap V_i| = (d \pm \varepsilon)|V_i|$ for every $i \in [2]$ and $v \in V_{3-i}$;
- *lower* (ε, d)-*regular* if $d_G(U, W) \geq d - \varepsilon$ for every $U \subseteq V_1, W \subseteq V_2$ satisfying $|U| > \varepsilon|V_1|$, $|W| > \varepsilon|V_2|$; and
- *lower* (ε, d)-*super-regular* if it is lower (ε, d)-regular and $|N_G(v) \cap V_i| \geq (d - \varepsilon)|V_i|$ for every $i \in [2]$ and $v \in V_{3-i}$.

Definition 2.4. We say that $\big(H, G, R, (X_i)_{i\in[r]}, (V_i)_{i\in[r]}\big)$ is a *blow-up instance* if

- H and G are graphs,
- $(X_i)_{i\in[r]}$ is a partition of $V(H)$ into independent sets,
- $(V_i)_{i\in[r]}$ is a partition of $V(G)$ satisfying $|X_i| \leq |V_i|$ for all $i \in [r]$, and
- R is a graph with $V(R) = [r]$ such that $H[X_i, X_j]$ is empty for all $ij \notin E(R)$.

In addition, for $\varepsilon > 0$ and $d \in (0, 1)$, we say the blow-up instance is (ε, d)-*regular*, *lower* (ε, d)-*regular*, (ε, d)-*super-regular*, or *lower* (ε, d)-*super-regular* if for every $ij \in E(R)$, the graph $G[V_i, V_j]$ is (ε, d)-regular, lower (ε, d)-regular, (ε, d)-super-regular, or lower (ε, d)-super-regular, respectively.

If $(H, G, R, (X_i)_{i\in[r]}, (V_i)_{i\in[r]})$ is a blow-up instance, then H embeds into the graph obtained by making $G[V_i, V_j]$ complete for every $ij \in E(R)$. The classical Blow-up Lemma states that the same conclusion holds when these complete pairs are replaced by super-regular pairs, provided that H has bounded maximum degree, the clusters have equal size, ε is sufficiently small, and the cluster size is sufficiently large. The Partial Embedding Lemma requires only lower regularity, but embeds a graph which is considerably smaller than the host graph.

The proof of the Partial Embedding Lemma also gives the vertex-spread form stated in Theorem 4.1. The main theorem gives the analogous strengthening of the Blow-up Lemma. We allow candidate sets for individual vertices using the following definition.

Definition 2.5. We say a bipartite graph A^i with bipartition (X_i, V_i) is a *candidacy graph* for (X_i, V_i), and we say $(H, G, R, (X_i)_{i\in[r]}, (V_i)_{i\in[r]}, (A^i)_{i\in[r]})$ is a *blow-up instance with candidacy graphs* if $(H, G, R, (X_i)_{i\in[r]}, (V_i)_{i\in[r]})$ is a blow-up instance and A^i is a candidacy graph for (X_i, V_i) for all $i \in [r]$. In this case, we say the blow-up instance with candidacy graphs is *lower* (ε, d)-*super-regular* if the blow-up instance is lower (ε, d)-super-regular and A^i is lower (ε, d)-super-regular for all $i \in [r]$. We also say an embedding $\varphi : H \hookrightarrow G$ is *compatible* with the candidacy graphs $A^1, \dots, A^r$ if for every $i \in [r]$ and $x \in X_i$ we have $\varphi(x) \in N_{A^i}(x)$.

We can now state the vertex-spread Blow-up Lemma.

Theorem 2.6. *For every d, Δ, r, there exist $\varepsilon = \varepsilon_{2.6}(d, \Delta)$, $C = C_{2.6}(d, \Delta)$ and $n_0 = n_{2.6}(d, \Delta, r)$ such that the following holds for all $n \geq n_0$. Let $\big(H, G, R, (X_i)_{i\in[r]}, (V_i)_{i\in[r]}, (A^i)_{i\in[r]}\big)$ be a lower (ε, d)-super-regular blow-up instance with candidacy graphs. If $\Delta(H), \Delta(R) \leq \Delta$ and $(1 - \varepsilon)n \leq |V_i| \leq (1 + \varepsilon)n$ for every $i \in [r]$, then there exists a (C/n)-vertex-spread distribution on embeddings of H into G that are compatible with $A^1, \dots, A^r$.*

We briefly describe the proof. Rödl and Ruciński [13] construct an embedding cluster by cluster. Since each X_i is independent, embedding X_i amounts to choosing a perfect matching in its candidacy graph. After each choice, the candidacy graphs for the unembedded clusters are updated, and the Four Graphs lemma shows that these graphs remain super-regular with high probability.

Our main probabilistic input is Theorem 5.1, which states that the uniform distribution on perfect matchings of a super-regular bipartite graph with parts of size n is $O(1/n)$-spread. Pham, Sah, Sawhney, and Simkin [12] proved that such a graph has some $O(1/n)$-spread distribution on perfect matchings, but here it is essential that the uniform distribution has this property. We prove this using a switching argument inspired by [9, Theorem 4.3]. Conditioning the uniform distribution on the high-probability event supplied by the Four Graphs lemma then gives a vertex-spread choice at each stage of the embedding.

The proof has three parts. In Section 5, we prove the perfect-matching spread result. In Section 6, we reduce the main theorem to an auxiliary form in which every pair of source clusters induces a saturating matching. Finally, in Section 7, we randomize the cluster-by-cluster embedding procedure to prove the auxiliary lemma. The reduction also handles unequal cluster sizes by first applying Theorem 4.1 to a small proportion of the vertices.

The statement of Theorem 2.6 is slightly stronger than the version of the Blow-up Lemma in [13]: the reduced graph need not be complete, candidacy graphs are allowed, the regularity parameter does not depend on the number of clusters, and the clusters need not all have the same size.

3. An alternate formulation

For applications, it is convenient to reformulate the main theorem using a homomorphism from the graph being embedded to the reduced graph.

Definition 3.1. A *homomorphism* from a graph H to a graph G, written $\phi : H \to G$, is a function $\phi : V(H) \to V(G)$ preserving adjacency; that is, $\phi(u)\phi(v) \in E(G)$ for every $uv \in E(H)$. Given a homomorphism $\phi : H \to G$, for any $S \subseteq V(G)$, we let $\phi^{-1}(S) \coloneqq \{x \in V(H) : \phi(x) \in S\}$, and for any $v \in V(G)$, we let $\phi^{-1}(v) \coloneqq \phi^{-1}(\{v\})$.

If R has no loops, a partition $(X_i)_{i\in[r]}$ of $V(H)$ into independent sets satisfies $E(H[X_i, X_j]) = \emptyset$ whenever $ij \notin E(R)$ if and only if the function $\phi : V(H) \to V(R)$ defined by $\phi(x) = i$ for $x \in X_i$ is a homomorphism.

Definition 3.2. Let G be a graph, let $(V_i)_{i\in[r]}$ be a partition of $V(G)$, and let R be a graph with $V(R) = [r]$. For $\varepsilon > 0$ and $d \in (0, 1)$, we say $(G, (V_i)_{i\in[r]})$ is an (ε, d)*-regular blow-up* of R if for every $ij \in E(R)$, the graph $G[V_i, V_j]$ is (ε, d)-regular. We define *lower* (ε, d)*-regular blow-ups*, (ε, d)*-super-regular blow-ups*, and *lower* (ε, d)*-super-regular blow-ups* similarly.

We also use a single candidacy graph to encode all candidate sets.

Definition 3.3. Let H and G be graphs. A *candidacy graph* for H and G is a bipartite graph with bipartition $(V(H), V(G))$. An embedding $\varphi : H \hookrightarrow G$ is *compatible* with a candidacy graph A if for every $x \in V(H)$, we have $\varphi(x) \in N_A(x)$.

The homomorphism formulation of the vertex-spread Blow-up Lemma is as follows.

Theorem 3.4. *For every d, Δ, r, there exist $\varepsilon = \varepsilon_{3.4}(d, \Delta)$, $C = C_{3.4}(d, \Delta)$ and $n_0 = n_{3.4}(d, \Delta, r)$ such that the following holds for all $n \geq n_0$. Let $(G, (V_i)_{i\in[r]})$ be a lower (ε, d)-super-regular blow-up of a graph R. Let H be a graph, and let A be a candidacy graph for H and G. Suppose $\Delta(H), \Delta(R) \leq \Delta$ and $(1 - \varepsilon)n \leq |V_i| \leq (1 + \varepsilon)n$ for every $i \in [r]$. If there is a homomorphism $\phi : H \to R$ such that $|\phi^{-1}(i)| \leq |V_i|$ and $A[\phi^{-1}(i), V_i]$ is lower (ε, d)-super-regular for every $i \in [r]$, then there is a (C/n)-vertex-spread distribution on embeddings of H into G that are compatible with A.*

Proof of Theorem 3.4 assuming Theorem 2.6. For each $i \in [r]$, let $X_i \coloneqq \phi^{-1}(i)$, let $A^i \coloneqq A[X_i, V_i]$, and note that $(H, G, R, (X_i)_{i\in[r]}, (V_i)_{i\in[r]}, (A^i)_{i\in[r]})$ is a lower (ε, d)-super-regular blow-up instance with candidacy graphs. Therefore, by Theorem 2.6, there is a (C/n)-vertex-spread distribution on embeddings of H into G that are compatible with A, as desired. □

4. Preliminaries

We use standard hierarchy notation. For example, a statement made under the assumption

$$1/n \ll a \ll b$$

means that for every fixed $b > 0$, there exists $a_0 > 0$ such that for every $a \in (0, a_0]$, there exists n_0 such that the statement holds for every $n \geq n_0$. Hierarchies with more terms are interpreted similarly, with the constants chosen from right to left. We write $x = y \pm z$ to mean $y - z \leq x \leq y + z$.

4.1. **Partial embeddings.** Let H and G be graphs, let A be a candidacy graph for H and G, and let $X \subseteq V(H)$. If $f : H[X] \hookrightarrow G$ is an embedding of $H[X]$ compatible with A, then we define the *updated candidacy graph* for A and f, denoted A_f, to be the subgraph of A with bipartition $(V(H) \setminus X, V(G) \setminus f(X))$, where for every $x \in V(H) \setminus X$ and $v \in V(G) \setminus f(X)$, we have $xv \in E(A_f)$ if $xv \in E(A)$ and $f(x')v \in E(G)$ for every $x' \in N_H(x) \cap X$. Note that if $f' : H - X \hookrightarrow G - f(X)$ is an embedding compatible with A_f, then f and f' can be combined to obtain an embedding $H \hookrightarrow G$.

If $\big(H, G, R, (X_i)_{i\in[r]}, (V_i)_{i\in[r]}, (A^i)_{i\in[r]}\big)$ is a blow-up instance with candidacy graphs, $X \subseteq V(H)$, and $f : H[X] \hookrightarrow G$ is an embedding of $H[X]$ compatible with $A^1, \dots, A^r$, then for each $i \in [r]$, we let $A^i_f \subseteq A_f$ be the subgraph induced by $X_i \setminus X$ and $V_i \setminus f(X)$.

Lemma 4.1. *Let $1/n \ll 1/C, \varepsilon \ll d' \ll d, 1/\Delta$, and let $\big(H, G, R, (X_i)_{i\in[r]}, (V_i)_{i\in[r]}, (A^i)_{i\in[r]}\big)$ be a lower (ε, d)-regular blow-up instance with candidacy graphs. If $\Delta(H), \Delta(R) \le \Delta$, $|X_i| \le \varepsilon|V_i|$ and $|V_i| = (1 \pm \varepsilon)n$ for every $i \in [r]$, and $\delta(A^i) \ge dn$ for every $i \in [r]$, then there exists a (C/n)-vertex-spread distribution on embeddings of H into G that are compatible with $A^1, \dots, A^r$. Moreover, for every $X \subseteq V(H)$, there exists a (C/n)-vertex-spread distribution on embeddings $f : H[X] \hookrightarrow G$ compatible with $A^1, \dots, A^r$ such that $\delta(A^i_f) \ge d'n$ for every $i \in [r]$ for which $X \subsetneq X_i$.*

Let $(G, (V_i)_{i\in[r]})$ be a lower (ε, d)-regular blow-up of a graph R. Let H be a graph, and let A be a candidacy graph for H and G. Suppose $\Delta(H), \Delta(R) \le \Delta$ and $|V_i| = (1 \pm \varepsilon)n$ for every $i \in [r]$. If there is a homomorphism $\phi : H \to R$ such that $|\phi^{-1}(i)| \le \varepsilon|V_i|$ for every $i \in [r]$ and $|N_A(x) \cap V_{\phi(x)}| \ge dn$ for every $x \in V(H)$, then there exists a (C/n)-vertex-spread distribution on embeddings of H into G that are compatible with A. Moreover, for every $X \subseteq V(H)$, there exists a (C/n)-vertex-spread distribution on embeddings $f : H[X] \hookrightarrow G$ compatible with A such that $|N_{A_f}(x) \cap V_{\phi(x)}| \ge d'n$ for every $x \in V(H) \setminus X$.

Note that the blow-up instance in Theorem 4.1 does not need to be super-regular because H is much smaller than G here. However, we do need a minimum-degree condition for the candidacy graphs. Theorem 4.1 is essentially the same as the "Partial Embedding Lemma" proved by Böttcher, Schacht, and Taraz [3, Lemma 10], with a few minor differences which do not affect the proof. The proof of that lemma is also essentially the same as that of the "Key Lemma" [10], implicit in the earlier work of Chvatál, Rödl, Szemerédi, and Trotter [4]. Here we only require the blow-up instance to be lower (ε, d)-regular, but the lemma is equivalent to the lemma with this condition replaced with the condition that the blow-up instance be (ε, d)-regular by Theorem 4.2. Here we also have candidacy graphs, but the argument is the same with candidacy graphs of minimum degree at least dn.

Theorem 4.1 can be proved by induction on $|X|$ by proving the slightly stronger statement that there exists a (C/n)-vertex-spread distribution on embeddings $f : H[X] \hookrightarrow G$ compatible with $A^1, \dots, A^r$ such that for every $i \in [r]$ and $x \in X_i \setminus X$, the vertex x has at least $(d-\varepsilon)^{1+|N_H(x)\cap X|} - |X|$ neighbors in the candidacy graph A^j_f. If $f : H[X \setminus \{x\}] \hookrightarrow G$ is an embedding from the inductive hypothesis (chosen randomly according to the (C/n)-vertex-spread distribution), then there are least $d'n/2$ choices for $f(x)$ with the desired property. Choosing one of these uniformly at random yields a probability distribution on embeddings $f : H[X] \hookrightarrow G$, which, by a standard application of the Chain rule of probability, is easily seen to be (C/n)-vertex-spread, as desired.

4.2. **Regularity.**

Lemma 4.2. *Let $1/n \ll \varepsilon \ll \varepsilon', d < 1$, and let G be a bipartite graph with bipartition $\{U, W\}$ where $|U| = |W| = n$.*

(a) If G is lower $(\varepsilon, d+\varepsilon)$-regular, then G contains a spanning subgraph which is (ε', d)-regular.

(b) If G is lower $(\varepsilon, d+\varepsilon)$-regular and $\delta(G) \ge dn$, then G contains a spanning subgraph which is $(\varepsilon', d^2/2)$-super-regular.

This is [13, Fact 2].

5. Perfect matchings in super-regular bipartite graphs

The main probabilistic ingredient is that the uniform distribution on perfect matchings of a super-regular bipartite graph is $O(1/n)$-spread.

Lemma 5.1. *Let $1/n \ll 1/C \ll \varepsilon \ll d$, and let G be a bipartite graph with parts of size n. If G is (ε, d)-super-regular, then the uniform distribution on perfect matchings of G is (C/n)-spread.*

In particular, Theorem 5.1 implies [12, Theorem 4.2], which asserts the existence of a (C/n)-spread distribution on perfect matchings in every such graph.

A well-known consequence of the results of Egorychev [5] and Falikman [6] is that every k-regular bipartite graph with parts of size n has at least $(k/n)^n n!$ perfect matchings. Alon, Rödl, and Ruciński [2] used this fact to prove the following estimate.

Lemma 5.2 (Alon, Rödl, and Ruciński [2])**.** *Let $1/n \ll \varepsilon \ll d$, and let G be a bipartite graph with parts of size n. If G is (ε, d)-super-regular, then G has at least $(d - 2\varepsilon)^n n!$ perfect matchings.*

The enumeration estimate gives the required bound when many edges are prescribed.

Lemma 5.3. *Let $1/n \ll 1/C \ll \varepsilon \ll d$, let G be an (ε, d)-super-regular bipartite graph with parts of size n, and let $e_1, \dots, e_s \in E(G)$ be distinct edges. If M is a perfect matching of G chosen uniformly at random, then*

$$\mathbb{P}[e_1, \dots, e_s \in M] \leq \frac{C^n}{n^s}.$$

Proof. By Theorem 5.2, G contains at least $(d - 2\varepsilon)^n n!$ perfect matchings. Moreover, there are at most $(n - s)!$ perfect matchings of G which contain $e_1, \dots, e_s$. Therefore,

$$\mathbb{P}[e_1, \dots, e_s \in M] \leq \frac{(n-s)!}{(d-2\varepsilon)^n n!}.$$

Since $n! \geq (n/e)^n$ and $(n-s)! \leq e(n-s)((n-s)/e)^{n-s} \leq en(n/e)^{n-s}$, we have

$$\frac{(n-s)!}{n!} \leq en\left(\frac{e}{n}\right)^s.$$

Combining the two inequalities above, we have

$$\mathbb{P}[e_1, \dots, e_s \in M] \leq \left(\frac{(en)^{1/n}}{d-2\varepsilon}\right)^n \left(\frac{e}{n}\right)^s \leq \frac{C^n}{n^s},$$

as desired. □

For small sets of prescribed edges, we instead use the chain rule together with the following switching estimate, inspired by [9, Theorem 4.3].

Lemma 5.4. *Let $1/n \ll 1/C \ll \varepsilon \ll d$, let G be an (ε, d)-super-regular bipartite graph with parts of size n, and let N be a matching in G of size at most $dn/2$. If M is a perfect matching of G chosen uniformly at random, then for every edge $e \in E(G) \setminus N$,*

$$\mathbb{P}[e \in M \mid N \subseteq M] \leq \frac{C}{n}.$$

Proof. We use a switching argument. Let $\mathcal{M}_{\text{in}}$ be the set of perfect matchings of G containing $N \cup \{e\}$, and let $\mathcal{M}_{\text{out}}$ be the set of perfect matchings of G containing N and not containing e. We assume without loss of generality there is at least one perfect matching in G containing $N \cup \{e\}$, or else $\mathbb{P}[e \in M \mid N \subseteq M] = 0$, so $\mathcal{M}_{\text{in}} \neq \emptyset$. Note that

$$\mathbb{P}[e \in M \mid N \subseteq M] = \frac{|\mathcal{M}_{\text{in}}|}{|\mathcal{M}_{\text{in}}| + |\mathcal{M}_{\text{out}}|}. \tag{5.1}$$

We will show that $|\mathcal{M}_{\text{out}}| \geq n|\mathcal{M}_{\text{in}}|/C$.

For every perfect matching M of G, if C is a 6-cycle in G containing three edges of M, then we say C is a *switch* of M. (The idea is that $M \triangle C$, the matching obtained by "switching" the edges of C, is also a perfect matching in G). We say a switch C is *valid* if C does not contain any edges of N. We construct an auxiliary bipartite graph A with bipartition $(\mathcal{M}_{\text{in}}, \mathcal{M}_{\text{out}})$ where $M_1M_2 \in E(A)$ if $M_1 = M_2 \triangle C$ for some valid switch C of M_1. Note that if $M_1 = M_2 \triangle C$ where $M_1 \in \mathcal{M}_{\text{in}}$ and $M_2 \in \mathcal{M}_{\text{out}}$, then C is a valid switch of both M_1 and M_2 and e is an edge of C.

First, we claim that

$$d_A(M) \leq n \text{ for every } M \in \mathcal{M}_{\text{out}}. \tag{5.2}$$

Indeed, for every $M \in \mathcal{M}_{\text{out}}$, every valid switch of M contains e and the two unique edges of M incident to the ends of e. Hence, every valid switch C of M is uniquely determined by the M-edge in C not incident to an end of e. Since there are at most n such edges, (5.2) follows.

Next we claim that

$$d_A(M) \geq (d-\varepsilon)(d/2-\varepsilon)^2 n^2 \text{ for every } M \in \mathcal{M}_{\text{in}}. \tag{5.3}$$

To that end, let $M \in \mathcal{M}_{\text{in}}$, let u and w be the ends of e, let $U := N_M(N_G(u) \setminus V(N))$, and let $W := N_M(N_G(w) \setminus V(N))$. That is, U is the set of vertices of G that are the ends of a length-2 $(M \setminus N)$-alternating path with u as the other end, and W is the set of vertices of G that are the ends of a length-2 $(M \setminus N)$ with w as the other end. Note that every edge of $G[U,W]$ corresponds to a distinct valid switch C of M. Since $|N| \leq dn/2$ and G is (ε, d)-super-regular, $|U| = |N_G(u) \setminus V(N)| \geq d_G(v) - |E(N)| \geq (d/2-\varepsilon)n$. Similarly, $|W| \geq (d/2-\varepsilon)n$. Hence, since G is (ε, d)-super-regular, we have $d_G(U,W) \geq d - \varepsilon$, and thus,

$$e_G(U,W) \geq (d-\varepsilon)|U||W| \geq (d-\varepsilon)(d/2-\varepsilon)^2 n^2.$$

Therefore there are at least $(d-\varepsilon)(d/2-\varepsilon)^2 n^2$ valid switches of M, and (5.3) follows.

By (5.2) and (5.3), we have

$$n^2|\mathcal{M}_{\text{in}}|/C \leq (d-\varepsilon)(d/2-\varepsilon)^2 n^2 |\mathcal{M}_{\text{in}}| \leq |E(A)| \leq n|\mathcal{M}_{\text{out}}|,$$

so $|\mathcal{M}_{\text{out}}| \geq n|\mathcal{M}_{\text{in}}|/C$. Therefore,

$$\frac{|\mathcal{M}_{\text{in}}|}{|\mathcal{M}_{\text{in}}| + |\mathcal{M}_{\text{out}}|} \leq \frac{1}{1 + n/C} \leq \frac{C}{n},$$

and the result follows from (5.1). □

Combining Theorems 5.3 and 5.4 proves the perfect-matching spread lemma.

Proof of Theorem 5.1. Let $e_1, \dots, e_s \in E(G)$ be distinct edges, and let M be a perfect matching of G chosen uniformly at random. We will show

$$\mathbb{P}[e_1, \dots, e_s \in M] \leq \left(\frac{C}{n}\right)^s.$$

We assume that $\{e_1, \dots, e_s\}$ is a matching, or else $\mathbb{P}[e_1, \dots, e_s \in M] = 0$. We consider two cases, depending on whether $s \geq dn/2$. If so, then we use Theorem 5.3, and otherwise, we use Theorem 5.4.

First, suppose $s \geq dn/2$. By Theorem 5.3 with $C^{d/2}$ playing the role of C, we have

$$\mathbb{P}[e_1, \dots, e_s \in M] \leq \frac{C^{nd/2}}{n^s} \leq \left(\frac{C}{n}\right)^s,$$

as desired.

Hence, we may assume $s < dn/2$. For each $i \in [s]$, let $N_i := \{e_1, \dots, e_i\}$, and let $N_0 = \emptyset$. By Theorem 5.4, for every $i \in [s]$,

$$\mathbb{P}[e_i \in M \mid N_{i-1} \subseteq M] \leq C/n.$$

Therefore, by the Chain Rule,

$$\mathbb{P}\left[e_1,\dots,e_s \in M\right] = \prod_{i=1}^{s} \mathbb{P}\left[e_i \in M \mid N_{i-1} \subseteq M\right] \le \left(\frac{C}{n}\right)^s,$$

as desired. □

6. Reduction to the auxiliary lemma

We reduce Theorem 2.6 to the following vertex-spread analogue of the auxiliary lemma of Rödl and Ruciński [13].

Lemma 6.1. *Let $1/n \ll \varepsilon, 1/C \ll d, 1/\Delta$ and $1/n \ll 1/r$. Let $(H, G, R, (X_i)_{i\in[r]}, (V_i)_{i\in[r]}, (A^i)_{i\in[r]})$ be an (ε, d)-super-regular blow-up instance with candidacy graphs. If*

- *$\Delta(H), \Delta(R) \le \Delta$,*
- *$|X_i| = |V_i| \in \{n, n+1\}$ for every $i \in [r]$, and*
- *$H[X_i, X_j]$ is a saturating matching for every $ij \in E(R)$ and empty for every $ij \notin E(R)$,*

then there exists a (C/n)-vertex-spread distribution on embeddings of H into G that are compatible with $A^1, \dots, A^r$.

We first consider the case in which $|V_1| = \cdots = |V_r| = n$. We refine the partitions of H and G so that every nonempty source pair is a saturating matching, following the reduction in [13]. The same refinement is applied to the candidacy graphs; see, for example, [7, Lemma 5.2]. The reduced graph need not be complete, since pairs corresponding to its nonedges play no role. Moreover, by Theorem 4.2, every lower super-regular pair contains a spanning super-regular subgraph. These observations give the following reduction lemma.

Lemma 6.2. *Let $1/n \ll \varepsilon \ll \varepsilon' \ll d' \ll d, 1/\Delta$ and $1/n \ll 1/r$. Let $\big(H, G, R, (X_i)_{i\in[r]}, (V_i)_{i\in[r]}, (A^i)_{i\in[r]}\big)$ be a lower (ε, d)-super-regular blow-up instance with candidacy graphs. If $\Delta(H), \Delta(R) \le \Delta$ and $|X_i| = |V_i| = n$ for every $i \in [r]$, then there exists an (ε', d')-super-regular blow-up instance with candidacy graphs $(H', G', R', (X_{i,j})_{i\in[r],j\in[\Delta^2]}, (V_{i,j})_{i\in[r],j\in[\Delta^2]}, (A^{i,j})_{i\in[r],j\in[\Delta^2]})$ where*

- *$H' \supseteq H$, $G' \subseteq G$, and $A^{i,j} \subseteq A^i$ for every $(i,j) \in [r] \times [\Delta^2]$,*
- *$X_i = \bigcup_{j=1}^{\Delta^2} X_{i,j}$ and $V_i = \bigcup_{j=1}^{\Delta^2} V_{i,j}$ for every $i \in [r]$,*
- *$\Delta(R') \le \Delta^3$,*
- *$|X_{i,j}| = |V_{i,j}| \in \{n', n'+1\}$, where $n' = \lfloor n/\Delta^2 \rfloor$, and*
- *$H'[X_{i,j}, X_{i',j'}]$ is a saturating matching for every $(i,j)(i',j') \in E(R')$ and empty otherwise.*

Proof. Let R' be the graph with $V(R') \coloneqq [r] \times [\Delta^2]$ where $(i,j)(i',j') \in E(R')$ if $ii' \in E(R)$, and note that $\Delta(R) \le \Delta^3$, as required. By applying the Hajnal–Szemerédi Theorem [8] to $H^2[X_i]$ for each $i \in [r]$, we can partition X_i into Δ^2 parts $X_{i,1}, \dots, X_{i,\Delta^2}$ of size in $\{n', n'+1\}$, where $n' \coloneqq \lfloor n/\Delta^2 \rfloor$, such that each part is a 2-independent set of H, meaning vertices in different parts are nonadjacent and have no common neighbors. In particular, $H[X_{i,j}, X_{i',j'}]$ is a matching if $(i,j)(i',j') \in E(R')$ and empty otherwise. By possibly adding edges to H to obtain $H' \supseteq H$, we have $H'[X_{i,j}, X_{i',j'}]$ is a saturating matching if $(i,j)(i',j') \in E(R')$ and empty otherwise, as desired. Using standard probabilistic arguments, we can partition each V_i accordingly into $V_{i,1}, \dots, V_{i,\Delta^2}$ so that the new blow-up instance $(H', G, R', (X_{i,j})_{i\in[r],j\in[\Delta^2]}, (V_{i,j})_{i\in[r],j\in[\Delta^2]}, (A^{i,j})_{i\in[r],j\in[\Delta^2]})$ with candidacy graphs $A^{i,j} \coloneqq A^i[X_{i,j}, V_{i,j}]$ is lower $(2\Delta^2\varepsilon, d)$-super-regular. Each candidacy graph $A^{i,j}$ and lower (ε', d)-super-regular subgraph $G[V_{i,j}, V_{i',j'}]$ contains a spanning (ε', d')-super-regular subgraph by Theorem 4.2. Let $(A')^{i,j} \subseteq A^{i,j}$ be such an (ε', d')-super-regular subgraph and let $G' \subseteq G$ be the union of these subgraphs for all $(i,j)(i',j') \in E(R)$. Now $(H', G', R', (X_{i,j})_{i\in[r],j\in[\Delta^2]}, (V_{i,j})_{i\in[r],j\in[\Delta^2]}, ((A')^{i,j})_{i\in[r],j\in[\Delta^2]})$ is the desired blow-up instance. □

It remains to equalize the cluster sizes. We use Theorem 4.1 to embed a small proportion of the vertices while preserving lower super-regularity of the remaining blow-up instance.

Lemma 6.3. *Let $1/n \ll \varepsilon, 1/C \ll \varepsilon' \ll d' \ll d, 1/\Delta$ and $1/n \ll 1/r$. Let $(H, G, R, (X_i)_{i\in[r]}, (V_i)_{i\in[r]}, (A^i)_{i\in[r]})$ be a lower (ε, d)-super-regular blow-up instance with candidacy graphs. If $\Delta(H), \Delta(R) \le \Delta$ and $(1-\varepsilon)n \le |X_i| = |V_i| \le (1+\varepsilon)n$ for every $i \in [r]$, then there exists $X \subseteq V(H)$ and a (C/n)-vertex-spread distribution on embeddings $f : H[X] \hookrightarrow G$ that are compatible with $A^1, \dots, A^r$ such that $(H-X, G-f(X), R, (X_i \setminus X)_{i\in[r]}, (V_i \setminus f(X))_{i\in[r]}, (A^i_f)_{i\in[r]})$ is a lower (ε', d')-super-regular blow-up instance with candidacy graphs such that $|V_i \setminus f(X)| = \lfloor (1-\varepsilon)n \rfloor$ for every $i \in [r]$.*

Proof. Let $X \subseteq V(H)$ satisfy $|X \cap X_i| = |V_i| - \lfloor (1-\varepsilon)n \rfloor$ for each $i \in [r]$, and let $X' = \bigcup_{x\in X} N_H(x) \setminus X$. Note that $|X \cap X_i| \le 3\varepsilon n$ and $|X' \cap X_i| \le 3\varepsilon\Delta^2 n$ for every $i \in [r]$. Hence, by Theorem 4.1 with $H[X']$ playing the role of H and $\Delta^2\varepsilon$ playing the role of ε, there exists a (C/n)-vertex-spread distribution on embeddings $f : H[X] \hookrightarrow G$ that are compatible with $A^1, \dots, A^r$ such that $\delta(A^i_f[X' \cup V_i]) \ge d'n$ for every $i \in [r]$ for which $X' \subsetneq X_i \cap X$.

It suffices to show that $(H-X, G-f(X), R, (X_i \setminus X)_{i\in[r]}, (V_i \setminus f(X))_{i\in[r]}, (A^i_f)_{i\in[r]})$ is a lower (ε', d')-super-regular blow-up instance with candidacy graphs such that $|V_i \setminus f(X)| = \lfloor (1-\varepsilon)n \rfloor$ for every $i \in [r]$ for each $f : H[X] \hookrightarrow G$ in the support of this distribution. Since $|X \cap X_i| = |V_i| - \lfloor (1-\varepsilon)n \rfloor$ for every $i \in [r]$, we have $|V_i \setminus f(X)| = \lfloor (1-\varepsilon)n \rfloor$ for every $i \in [r]$, as desired. Since $\delta(A^i_f[X' \cup V_i]) \ge d'n$ for every $i \in [r]$ for which $X' \subsetneq X_i \cap X$, $|X \cap X_i| = |V_i \cap f(X)| \le 3\varepsilon n$, and $|X' \cap X_i| \le 3\varepsilon\Delta^2 n$, the blow-up instance with candidacy graphs is lower (ε', d')-super-regular, as desired. □

We now combine the auxiliary lemma with Theorems 6.2 and 6.3.

Proof of Theorem 2.6 assuming Theorem 6.1. Let ε_1, ε_2, d_1, d_2, and C' satisfy $1/C' \ll \varepsilon, 1/C \ll \varepsilon_1 \ll \varepsilon_2 \ll d_2 \ll d_1 \ll d$.

By Theorem 6.3 with ε_1 and d_1 playing the roles of ε' and d', respectively, there exists $X \subseteq V(H)$ and a (C/n)-vertex-spread distribution on embeddings $f : H[X] \hookrightarrow G$ that are compatible with $A^1, \dots, A^r$ such that $(H-X, G-f(X), R, (X_i \setminus X)_{i\in[r]}, (V_i \setminus f(X))_{i\in[r]}, (A^i_f)_{i\in[r]})$ is a lower (ε_1, d_2)-super-regular blow-up instance with candidacy graphs such that $|V_i \setminus f(X)| = \lfloor (1-\varepsilon)n \rfloor$ for every $i \in [r]$.

For each f in the support of this distribution, we apply Theorem 6.2 to $(H-X, G-f(X), R, (X_i \setminus X)_{i\in[r]}, (V_i \setminus f(X))_{i\in[r]}, (A^i_f)_{i\in[r]})$ to obtain an (ε_2, d_2)-super-regular blow-up instance with candidacy graphs $(H', G', R', (X_{i,j})_{i\in[r], j\in[\Delta^2]}, (V_{i,j})_{i\in[r], j\in[\Delta^2]}, (A^{i,j})_{i\in[r], j\in[\Delta^2]})$ where

- $H' \supseteq H - X$, $G' \subseteq G - f(X)$, and $A^{i,j} \subseteq A^i_f$ for every $(i,j) \in [r] \times [\Delta^2]$,
- $X_i \setminus X = \bigcup_{j=1}^{\Delta^2} X_{i,j}$ and $V_i \setminus f(X) = \bigcup_{j=1}^{\Delta^2} V_{i,j}$ for every $i \in [r]$,
- $\Delta(R') \le \Delta^3$,
- $|X_{i,j}| = |V_{i,j}| \in \{n', n'+1\}$, where $n' = \lfloor \lfloor (1-\varepsilon)n \rfloor / \Delta^2 \rfloor$, and
- $H'[X_{i,j}, X_{i',j'}]$ is a saturating matching for every $(i,j)(i',j') \in E(R')$ and empty otherwise,

We apply Theorem 6.1 to this blow-up instance with ε_2, C', d_2, Δ^3, and $r\Delta^2$ playing the role of ε, C, d, Δ, and r, respectively, to obtain a (C'/n')-vertex-spread distribution on embeddings $H' \hookrightarrow G'$ compatible with the candidacy graphs $A^{i,j}$. Since $H' \supseteq H$ and $G' \subseteq G$ and $C'/n' \le C/n$, this distribution is also a (C/n)-vertex-spread distribution on embeddings $f' : H - X \hookrightarrow G - f(X)$ compatible with the candidacy graphs A^i_f.

We can combine these distributions to obtain a distribution on embeddings $H \hookrightarrow G$, which is (C/n)-vertex-spread by the Chain Rule of probability. □

7. Proof of the auxiliary lemma

We prove Theorem 6.1 by showing that the randomized embedding procedure of Rödl and Ruciński is (C/n)-vertex-spread. We first recall their Four Graphs lemma [13].

Let W_1, W_2, and W_3 be disjoint sets of size n. For every $1 \le i < j \le 3$, let F_{ij} be an (ε, d_{ij})-super-regular bipartite graph with bipartition (W_i, W_j). We say that (F_{12}, F_{13}, F_{23}) is $(\varepsilon, d_{12}, d_{13}, d_{23})$-*regular* if

$$|N_{F_{13}}(v) \cap N_{F_{23}}(u)| = (d_{13}d_{23} \pm \varepsilon)n \text{ for every } uv \in E(F_{12}). \tag{7.1}$$

For a perfect matching $\sigma : W_1 \to W_2$ in F_{12}, let A_σ be the spanning subgraph of F_{13} in which $v_1v_3 \in E(A_\sigma)$ if $\sigma(v_1)v_3 \in E(F_{23})$.

Lemma 7.1 (Rödl and Ruciński [13], Four Graphs lemma)**.** *Let $1/n \ll a \ll \varepsilon \ll \varepsilon' \ll d_{12}, d_{13}, d_{23}$, and let (F_{12}, F_{13}, F_{23}) be an $(\varepsilon, d_{12}, d_{13}, d_{23})$-regular triplet of graphs. If σ is a perfect matching of F_{12} chosen uniformly at random, then A_σ is $(\varepsilon', d_{13}d_{23})$-super-regular with probability at least $1 - (1-a)^n$.*

Author's correction: the colour-round induction. The following argument replaces the former minimum-exposure ordering and the exposure-indexed error hierarchy in the proof of Theorem 6.1. The statement of the lemma is unchanged. The correction uses a bounded number of rounds obtained by colouring the square of the reduced graph. This colouring is unrelated to the equitable colouring of $H^2[X_i]$ used earlier to refine the source clusters.

We first record the two standard robust forms of the regularity inputs that we use. They are stated for parts differing by at most one because this is the only unequal-size case needed below.

Lemma 7.2 (Quantitative spanning thinning)**.** *Let $0 < \rho < D \le 1$ and $0 < \gamma < \rho$. There are $\beta_0 > 0$ and n_* such that the following holds. Let L, R be disjoint sets, put $s := \min\{|L|, |R|\}$, and suppose that $s \ge n_*$ and $||L| - |R|| \le 1$. If $F[L, R]$ is (β, D)-super-regular for some $0 < \beta \le \beta_0$, then it contains a spanning (γ, ρ)-super-regular subgraph. All degree conditions are normalised by the actual opposite-part size.*

Proof. Put $q := \rho/D$. Choose $\xi > 0$ and then $0 < \beta_0 \le \gamma$ so that $q\beta_0 + \xi \le \gamma$. Retain each edge independently with probability q. For every vertex, a Chernoff bound shows that its retained degree differs from q times its old degree by at most ξ times the opposite-part size, except with probability $\exp(-\Omega(s))$. If $U \subseteq L$ and $W \subseteq R$ are above the γ-thresholds, the retained edge count differs from $qe_F(U, W)$ by at most $\xi|U||W|$, except with probability $\exp(-\Omega(|U||W|)) = \exp(-\Omega(s^2))$. There are only $2^{|L|+|R|} = \exp(O(s))$ choices of (U, W). A union bound over all vertices and all such pairs is therefore smaller than one when s is large. Fix an outcome outside this union. Since $\beta \le \gamma$, old regularity applies to every tested (U, W), and

$$d_{F'}(U, W) = q\, d_F(U, W) \pm \xi = \rho \pm (q\beta + \xi).$$

The same calculation for every degree proves the result. □

Lemma 7.3 (Near-equal Four Graphs lemma)**.** *Fix $0 < d_{12}, d_{13}, d_{23} \le 1$, and put $q := d_{13}d_{23}$. For every sufficiently small $0 < \varepsilon' < \min\{q, 1\}$ there are $\varepsilon_0 > 0$, $0 < a \le 1$, and m_0 such that the following holds. Let $m \ge m_0$, let*

$$|W_1| = |W_2| = m, \qquad |W_3| = k \in \{m-1, m, m+1\},$$

and suppose that F_{12}, F_{13}, F_{23} are respectively (ε, d_{12})-, (ε, d_{13})-, and (ε, d_{23})-super-regular on the corresponding pairs, where $0 < \varepsilon \le \varepsilon_0$. Degrees are normalised by the actual opposite-part size. Suppose also that every $vu \in E(F_{12})$, with $v \in W_1$ and $u \in W_2$, satisfies

$$\left|N_{F_{13}}(v) \cap N_{F_{23}}(u) \cap W_3\right| = (q \pm \varepsilon)k. \tag{B1}$$

Then F_{12} has a perfect matching. If $\sigma : W_1 \to W_2$ is chosen uniformly from its perfect matchings, let A_σ be the graph on (W_1, W_3) in which

$$vw \in E(A_\sigma) \quad \iff \quad vw \in E(F_{13}) \text{ and } \sigma(v)w \in E(F_{23}).$$

Then

$$\Pr\big[A_\sigma \text{ is } (\varepsilon', q)\text{-super-regular}\big] \geq 1 - \big(1-a\big)^m. \tag{B2}$$

The constants are uniform over the three permitted values of k.

Proof. We indicate why the usual proof of the Four Graphs lemma is uniform in this range of sizes. Its matching set-image estimate says that, for fixed $d_{12}, \lambda, \delta > 0$, there are $c < 1$, an admissible input-error bound, and a size threshold such that, whenever $S \subseteq W_1$ and $T \subseteq W_2$ have size at least λm, a uniformly random perfect matching σ of F_{12} satisfies

$$\Pr\left[\left| |\sigma(S) \cap T| - \frac{|S||T|}{m} \right| > \delta m\right] \leq c^m. \tag{B3}$$

This is the standard two-edge-switching estimate used in the proof of the Four Graphs lemma [13]; its constants do not depend on S or T. Choose $0 < \delta \ll \varepsilon' \ll q, d_{12}, d_{13}, d_{23}$ and then choose ε_0 below the finitely many input bounds in (B3), using lower set-size bounds smaller than

$$d_{13}/2, \quad d_{23}/2, \quad (d_{13}/2)^2, \quad (d_{23}/2)^2.$$

For these choices, super-regularity of F_{12} also guarantees a perfect matching.

For $v \in W_1$, (B1) applied to $v\sigma(v)$ gives deterministically

$$\deg_{A_\sigma}(v) = (q \pm \varepsilon)k. \tag{B4}$$

For $w \in W_3$, set $S_w = N_{F_{13}}(w) \cap W_1$ and $T_w = N_{F_{23}}(w) \cap W_2$. These sets have sizes $(d_{13} \pm \varepsilon)m$ and $(d_{23} \pm \varepsilon)m$, while $\deg_{A_\sigma}(w) = |\sigma(S_w) \cap T_w|$. Thus (B3), followed by a union bound over $k \leq m+1$ vertices, gives

$$\deg_{A_\sigma}(w) = (q \pm O(\varepsilon + \delta))m \tag{B5}$$

simultaneously for all w, except with probability at most $(m+1)c^m$.

The standard good-codegree consequence of regularity says that all but $O(\varepsilon)k^2$ ordered pairs $(w, w') \in W_3^2$ have their two common neighbourhoods in W_1 and W_2 of sizes $(d_{13}^2 \pm O(\varepsilon))m$ and $(d_{23}^2 \pm O(\varepsilon))m$. Apply (B3) to these two sets and take a union bound over at most $(m+1)^2$ pairs. Except with additional probability at most $(m+1)^2 c^m$, every such pair has

$$|N_{A_\sigma}(w) \cap N_{A_\sigma}(w')| = (q^2 \pm O(\varepsilon + \delta))m. \tag{B6}$$

Choose δ and then ε_0 sufficiently small in terms of ε'. The degree–codegree criterion, applied with the actual part sizes, turns equations (B4)–(B6) into (ε', q)-super-regularity. Finally,

$$\big((m+1) + (m+1)^2\big)c^m \leq \big(1-a\big)^m$$

for some $a > 0$ and every sufficiently large m. The proof used only $m - 1 \leq k \leq m + 1$, which proves the asserted uniformity. □

Corrected proof of Theorem 6.1. Put $Q \coloneqq \Delta^2 + 1$. Let $R^{\langle 2 \rangle}$ be the graph on $V(R)$ in which distinct vertices are adjacent if they are adjacent in R or have a common neighbour in R. Since

$$\Delta(R^{\langle 2 \rangle}) \leq \Delta + \Delta(\Delta - 1) = \Delta^2,$$

there is a proper colouring

$$c : V(R) \longrightarrow \{0, \dots, Q-1\}.$$

We allow colours which are not used, and write $B_t \coloneqq c^{-1}(t)$. Thus two clusters in the same B_t are nonadjacent in R and have disjoint neighbourhoods in R. In particular, each cluster outside B_t has at most one neighbour in B_t.

We first fix the density parameters recursively. Let $p_0 = d$. For $0 \le x < \Delta$, having defined p_x, choose $0 < \rho_x < p_x$ and put

$$p_{x+1} \coloneqq \rho_x d/2.$$

Finally choose $0 < \rho_\Delta < p_\Delta$. For example, one may take $\rho_x = p_x/2$ at every step. The index x records the number of already embedded neighbours of a cluster. Our invariant stores a spanning super-regular graph of density p_x contained in the true updated candidacy graph; it does not assert that the whole updated candidacy graph has exactly that density.

Next choose error parameters

$$0 < \eta_0 \ll \eta_1 \ll \cdots \ll \eta_Q \ll d, 1/\Delta$$

from right to left. The choice is finite and depends only on d and Δ. Here is the precise order of choice. After η_{t+1} has been chosen, choose $\omega_{t+1} > 0$ below η_{t+1} and below the minimum of the finitely many admissible output-error caps in Theorem 7.3 for the density triples used below. For every ordered pair

$$(x, y) \in \{0, \dots, \Delta\} \times \{0, \dots, \Delta - 1\},$$

the near-equal Four Graphs lemma, with output error ω_{t+1} and density triple $(\rho_x, \rho_y, d/2)$, supplies a positive admissible input cutoff. Choose one common $\theta_t > 0$ below all of these cutoffs and also below the perfect-matching existence and uniform-spread cutoffs for the finitely many densities ρ_x. Instantiate Four Graphs at input θ_t, and denote its exponential constant for (x, y) by $a_{t,x,y}$. Likewise, let $M_{t,x}$ be a constant for the uniform-perfect-matching spread bound at density ρ_x. Thus

$$0 < a_{t,x,y} \le 1 \quad \text{and} \quad M_{t,x} \ge 1.$$

Only after these choices do we choose a post-thinning and padding error $\zeta_t > 0$ so small that

$$2\sqrt{\zeta_t} \le \theta_t, \qquad \zeta_t + \Delta\sqrt{\zeta_t} \le \theta_t, \qquad \zeta_t + 2\Delta\zeta_t \le \theta_t, \tag{A1}$$

and so that

$$2\zeta_t < \min\left\{ \min_{0 \le x \le \Delta} \rho_x, \frac{d}{2}, \min_{0 \le x \le \Delta} (1 - \rho_x), 1 - \frac{d}{2} \right\}. \tag{A2}$$

Choose an inner reserve $0 < \widehat{\zeta}_t < \zeta_t$. Finally choose η_t sufficiently small that every required spanning thinning of an (η_t, p_x)-super-regular witness has error at most $\widehat{\zeta}_t$, and also so that

$$2\eta_t < \min_{0 \le y < \Delta} (p_y - p_{y+1}).$$

It is important here to use all relevant ordered pairs (x, y): the colour-round order gives no inequality between the exposure of the current cluster and that of a future cluster. Taking the minimum of the finitely many $a_{t,x,y}$ and the maximum of the finitely many $M_{t,x}$ gives constants $0 < a \le 1$ and $M \ge 1$ which depend only on d and Δ. We take $C \ge 2M$. After r is fixed, choose n_0 last. Besides all thinning, row-padding, perfect-matching existence, uniform-spread, and Four Graphs thresholds, it is large enough that, for every $t < Q$ and $n \ge n_0$,

$$\Delta\big(1 - a\big)^n \le \tfrac{1}{2}, \qquad C/n \le 1. \tag{A3}$$

and

$$\omega_{t+1} + 2/n \le \eta_{t+1}, \qquad \omega_{t+1} + 1/(\eta_{t+1} n) \le \eta_{t+1}. \tag{A4}$$

The row insertion and integer-rounding errors are required to be at most $\zeta_t - \widehat{\zeta}_t$. Finally choose the input regularity parameter in Theorem 6.1 at most η_0, with η_0 also below every host-pair thinning cutoff from density d to $d/2$ in all Q rounds.

We embed the colour classes $B_0, \dots, B_{Q-1}$ in this order and process the clusters inside one colour class sequentially. For $S \subseteq B_t$, let

$$P_{t,S} \coloneqq \bigcup_{s<t} B_s \cup S$$

be the set of clusters already processed. If f embeds the corresponding source clusters and $j \notin P_{t,S}$, write A_f^j for the spanning subgraph of A^j in which xv is an edge exactly when it was an edge of A^j and $f(u)v \in E(G)$ for every already embedded neighbour u of x. The within-round invariant is the following. There is a (C/n)-vertex-spread distribution on compatible partial embeddings f of the source clusters indexed by $P_{t,S}$ such that, for every unembedded cluster j, if

$$x_j \coloneqq |N_R(j) \cap P_{t,S}|,$$

then the true updated candidacy graph A_f^j contains a spanning witness of density p_{x_j} which is

$$(*) \qquad \begin{cases} (\eta_{t+1}, p_{x_j})\text{-super-regular}, & \text{if } N_R(j) \cap S \neq \varnothing, \\ (\eta_t, p_{x_j})\text{-super-regular}, & \text{if } N_R(j) \cap S = \varnothing. \end{cases}$$

The first intersection in $(*)$ has size at most one. This touched/untouched distinction is essential: a witness changed earlier in round t is only required to have error η_{t+1}.

The invariant holds at the start with $t = 0$ and $S = \varnothing$, using the original candidacy graphs. Suppose it holds for $S \subseteq B_t$, and choose $i \in B_t \setminus S$. Since B_t is independent in $R^{\langle 2 \rangle}$, we have

$$N_R(i) \cap S = \varnothing \quad \text{and} \quad N_R(j) \cap S = \varnothing \quad \text{for every } j \in N_R(i) \setminus P_{t,S}.$$

Consequently, the witness for the current cluster i and every future witness which is tested while embedding i all have the common input error η_t, although their densities and exposure indices may differ.

Fix an outcome of the partial embedding up to this point. Apply Theorem 7.2 to thin the current witness of exposure x_i to density ρ_{x_i}, each relevant future witness of exposure x_j to density ρ_{x_j}, and each host pair $G[V_i, V_j]$ to density $d/2$, initially with error at most $\widehat{\zeta}_t$. Retain from the thinned current witness only those edges which satisfy the required degree-codegree test for every unembedded $j \in N_R(i)$, with the window

$$\big(\rho_{x_j} d/2 \pm 2\zeta_t\big)|V_j|$$

normalised by the actual future part. We spell out the simultaneous good-edge estimate, since it is the point at which the common error η_t is needed. After thinning, all pairs have error at most ζ_t. For one test and one fixed endpoint of the current pair, regularity of the two tested pairs shows that at most $2\zeta_t$ times the opposite part size fail the codegree test. Thus at most $2\Delta\zeta_t$ of the degree can be lost. If U and W are subsets of the two current parts of relative size greater than θ_t, the standard bad-pair count shows that at most a $\sqrt{\zeta_t}$ proportion of $U \times W$ fails any one test. Hence the total density loss on $U \times W$ is at most $\Delta\sqrt{\zeta_t}$. The three inequalities in (A1), together with regularity after a bounded density loss, show that the retained current pair is (θ_t, ρ_{x_i})-super-regular. They also show that it and each thinned future witness and host pair form the appropriate near-equal regular triplet. This proves the full super-regular conclusion needed here, rather than only the edgewise codegree condition.

By the perfect-matching existence threshold built into the hierarchy, the retained current pair has a perfect matching. Choose one uniformly. Its law is $(M/|X_i|)$-spread and therefore (M/n)-spread. For each unembedded $j \in N_R(i)$, the Four Graphs conclusion fails with probability at most $\big(1-a\big)^{|X_i|} \leq \big(1-a\big)^n$. By (A3), a union bound shows that all of the at most Δ local conclusions hold simultaneously with probability at least $1/2$. We condition this one matching on that local success event. The unconditioned matching law is (M/n)-spread, so the conditioned law is $(2M/n)$-spread and hence (C/n)-spread. For each such j we have $x_j < \Delta$, because i is an unembedded neighbour of j. Transport the Four Graphs output along the saturating source matching to obtain, inside the true updated candidacy graph, an (η_{t+1}, p_{x_j+1})-super-regular witness. Every other unembedded witness is unchanged. This proves the invariant with $S \cup \{i\}$ in place of S.

The conditioning in the preceding paragraph is local to the current cluster. We do not condition on one global event for all of B_t, whose probability could depend on $|B_t|$ and hence on r. Instead, the clusters of B_t are processed sequentially. At every step, conditional on the entire past, the new matching law is (C/n)-spread. The chain rule therefore preserves the joint (C/n)-vertex-spread bound. At the end of round t, every witness touched in that round already has error η_{t+1}, while every untouched witness has the stronger error η_t and may be viewed as having error η_{t+1}. This is precisely the invariant at the start of round $t+1$.

For completeness, we record the near-equal transport used above. Write $m = |X_i| = |V_i|$ and $k = |X_j| = |V_j|$, so $m, k \in \{n, n+1\}$ and $|m-k| \le 1$. The near-equal Four Graphs statement is normalised using the actual part sizes. If $m = k$, the output is transported directly along the perfect source matching. If $m = k+1$, the source matching leaves one current source vertex unmatched. Adjoin one artificial row to the pulled-back F_{13} pair: choose any subset of V_j whose size is in the required $(\rho_{x_j} \pm \zeta_t)k$ window. Adjoining this single row preserves super-regularity with the reserved slack. Regularity of the thinned host pair shows that all but at most $2\zeta_t m$ possible partners of this artificial row pass this future-neighbour test, so the simultaneous good-edge calculation above still applies to the entire current pair. We apply Four Graphs and then delete the matching-induced row corresponding to the unmatched current vertex.

If $k = m+1$, the saturating matching leaves one future source vertex z unmatched. Since $H[X_i, X_j]$ is a matching, z has no neighbour in X_i; its row in the true candidacy graph is therefore unchanged when X_i is embedded. Apply Four Graphs to the pullback along the saturating matching and transport its output first to $X_j \setminus \{z\}$. The stored old witness already supplies the row of z, of degree at least $(p_{x_j} - \eta_t)|V_j|$. Our parameter and size choices make this larger than $p_{x_j+1}|V_j| + 1$, since $p_{x_j+1} = \rho_{x_j} d/2 < p_{x_j}$. Choose a subset of that row's neighbourhood whose size is the nearest integer to $p_{x_j+1}|V_j|$, and adjoin precisely this row to the Four Graphs output. The result remains a subgraph of the true updated candidacy graph.

In either unequal case, the row-stability calculation is explicit: deleting or adjoining one row to an $(\omega_{t+1}, p_{x_j+1})$-super-regular pair preserves the degree conditions provided $\omega_{t+1} + 2/n \le \eta_{t+1}$, and preserves regularity provided $\omega_{t+1} + 1/(\eta_{t+1} n) \le \eta_{t+1}$. These are exactly (A4). The probabilistic estimates remain uniform because

$$M/m \le M/n \quad \text{and} \quad \big(1-a\big)^m \le \big(1-a\big)^n.$$

This proves the required one-row stability in both unequal-size cases and keeps every transported witness inside the true candidacy graph.

After the last colour round every cluster has been embedded. Repeated use of the chain rule gives a (C/n)-vertex-spread distribution on full compatible embeddings of H into G. This proves Theorem 6.1 with the same statement and quantifier order as before. No minimum-exposure ordering of the clusters is used. □

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